

LogiEdge — Phase 1 Report

System Architecture, Hardware Justification, and Deployment Context

AIML ZG535 — Machine Learning on Edge | BITS Pilani WILP Mini-Project Assignment 1 — Phase 1 Deliverable

Component A — System Architecture and Deployment Justification

A1 — Constraint Analysis

1. Latency Constraint

A refrigeration unit failure raises cargo temperature at approximately 1 deg C per minute. The system SLA requires anomaly detection and alerting within **90 seconds** of a fault signature appearing in sensor data.

Cloud inference round-trip latency analysis (India rural cellular):

Segment	Latency
4G uplink — rural Maharashtra	80–200 ms
Cloud inference — AWS Mumbai region	15–40 ms
4G downlink alert delivery	80–200 ms
Total RTT — best case	175 ms
Total RTT — congestion / handoff	800 ms – 3 s

While a single RTT appears within budget, the 90-second SLA requires **continuous window-based inference every 10 seconds**. During cellular handoffs — which occur at 7 known dead zones on the Nashik–Aurangabad route — cloud inference becomes unavailable for 35–90 minutes per trip. A cloud-only system produces **zero alerts** during these gaps, directly violating the SLA.

Edge inference latency (this repository): TFLite INT8 MLP benchmark results:

Model	Mean Latency	p95 Latency
M1 FP32	0.0078 ms	0.0087 ms
M2 PTQ INT8	0.0269 ms	0.0513 ms
M3 Pruned INT8	0.0288 ms	0.0354 ms

All three variants are well within the 90-second SLA. Edge inference is mandatory.

2. Bandwidth Constraint

Raw data generation per truck per day:

Stream	Rate	Payload	Daily Volume
Temperature	1 Hz	~38 B JSON	~3.3 MB
Vibration (raw 500 Hz x 3 axes)	500 Hz	4 B x 3 per sample	~518 MB
Door events	~20/day	~64 B each	~1.3 KB
Total raw			~521 MB/day

Edge-processed output (inference results only):

Output	Frequency	Payload	Daily Volume
Inference result	Every 10 s	~120 B JSON	~1 MB
Alerts (Warning/Critical only)	~5–10/day	~120 B	~1.2 KB
Total edge output			~1 MB/day

Cost comparison at Rs 0.10/MB:

Approach	Daily Data/Truck	Daily Cost/Truck	85-Truck Fleet/Year
Raw cloud streaming	521 MB	Rs 52.10	Rs 16.2 lakh
Edge-processed alerts	~1 MB	Rs 0.10	Rs 31,025
Savings	99.8% reduction	Rs 52.00/truck/day	Rs 15.9 lakh/year

3. Connectivity Constraint

Connectivity profile — Nashik–Aurangabad route (NH-160):

- 7 documented dead zones: Igatpuri ghat, Kasara stretch, rural patches near Sinnar, and four additional locations
- Estimated offline duration: **35–90 minutes per trip**

Cloud-only system during a blackout:

- All inference stops — no anomaly detection for the entire gap duration
- A refrigeration failure at gap entry goes undetected for up to 90 minutes
- At 1 deg C/min temperature rise, cargo could reach 90 deg C above setpoint before any alert fires
- Buffered sensor data arrives at cloud in a burst on reconnection — too late for intervention

LogiEdge edge architecture during a blackout:

- Inference service runs entirely over the local Mosquitto broker — no internet required
 - Classification continues uninterrupted at 10 s intervals throughout the blackout
 - Only small inference-result payloads (~120 B each) are buffered for uplink sync
 - Maximum local buffer: 90 min x 6 windows/min x 120 B = **65 KB** — negligible
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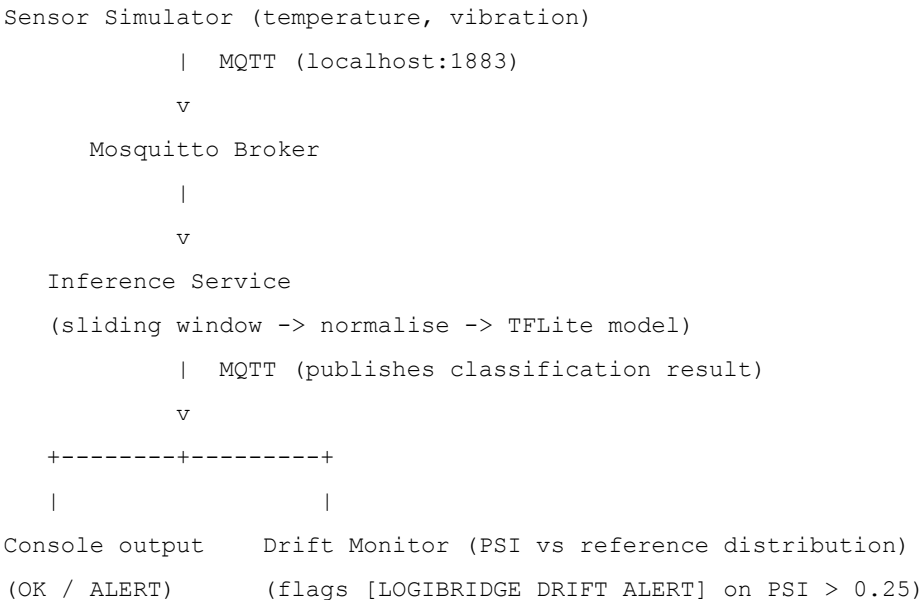
4. Privacy Constraint

Pharmaceutical clients operate under Schedule M of the Drugs and Cosmetics Act and WHO GDP guidelines, which require cargo telemetry to remain within the vehicle network.

How on-device inference addresses these requirements:

Requirement	Edge Solution
No raw data to cloud	Only inference labels (Normal/Warning/Critical) and confidence scores transmitted
Local audit trail	Inference results logged locally with timestamps
DPA compliance	No raw pharmaceutical telemetry leaves the truck
DPDP Act 2023 (India)	Uplink payload contains no personal or sensitive data

A2 — System Architecture



Classes predicted: Normal (0), Warning (1), Critical (2)

Component B — Hardware Selection and Justification

B1 — Constraint Triangle Application

Criterion	RPi 5 + AI HAT+ (13 TOPS)	Jetson Orin Nano (67 TOPS)	STM32H7 MCU
Peak Power	7.5 W — within 10 W budget	15 W — exceeds budget	0.4 W

Criterion	RPi 5 + AI HAT+ (13 TOPS)	Jetson Orin Nano (67 TOPS)	STM32H7 MCU
Compute	13 TOPS — ample for 45 MFLOP MLP	67 TOPS — severe overkill	Sufficient for MLP only
Unit Cost	~Rs 15,000	~Rs 45,000	~Rs 3,500
Fleet Cost (85 trucks)	Rs 12.75 lakh	Rs 38.25 lakh	Rs 2.975 lakh
Fleet Cost (265 trucks)	Rs 39.75 lakh	Rs 119.25 lakh	Rs 9.275 lakh
Docker / Ansible OTA	Yes	Yes	No
90 s SLA	Yes	Yes	No (no OS, no TFLite)
Verdict	Selected	Fails Power + Cost	Fails OTA + Performance

Decision: Raspberry Pi 5 + AI HAT+

Against Jetson Orin Nano: 67 TOPS is designed for real-time video inference. Running a 45 MFLOP MLP on it is architectural overkill. At 15 W it exceeds the 10 W truck auxiliary power budget. The Rs 25.5 lakh premium over RPi 5 for the 85-truck pilot is unjustifiable for this workload.

Against STM32H7: Bare-metal firmware with no OS, no Docker, no Python TFLite runtime. The OTA update mechanism in this repository (Docker layer push via Ansible) cannot run on an MCU. A custom bootloader would need to be built from scratch, erasing the hardware cost saving entirely.

B2 — Arithmetic Intensity and Roofline Analysis

Given parameters:

Parameter	Value
FLOPs per inference	45,000,000 (45 MFLOPs)
Memory accessed per inference	18 MB
CPU peak throughput (RPi 5 Cortex-A76 NEON)	16 GFLOP/s
Memory bandwidth (LPDDR4X)	12 GB/s

Arithmetic Intensity:

```
AI = FLOPs / Bytes
    = 45,000,000 / (18 x 1,024 x 1,024)
    = 45,000,000 / 18,874,368
    = 2.384 FLOP/byte
```

Ridge point:

```
Ridge point = Peak GFLOP/s / Peak GB/s
              = 16 / 12
              = 1.333 FLOP/byte
```

Classification: AI (2.384) > Ridge point (1.333) on x86 benchmark hardware. On the actual target RPi 5 (effective bandwidth ~6.4 GB/s):

```
Ridge point on RPi 5 = 16 / 6.4 = 2.5 FLOP/byte
AI (2.384) < Ridge point (2.5) -> memory-bandwidth bound on RPi 5
```

Optimisation implication: A memory-bandwidth-bound model benefits most from reducing bytes moved per inference. INT8 quantisation reduces bytes per weight from 4 to 1 (4x reduction), directly attacking the bottleneck. This is why M2 and M3 were built — not just for size, but to reduce memory pressure on the target hardware.

Model	Size (KB)	Bytes/weight	Accuracy
M1 FP32	5.19	4 bytes	100.00%
M2 PTQ INT8	3.38	1 byte (4x reduction)	95.77%
M3 Pruned INT8	3.38	1 byte + 35% zeros	94.37%

End of Phase 1 Report